

Snapmaker / 10W Laser Module / Laser Engraving and Cutting

The Definitive Guide to Laser Engraving and Cutting with the 10W High Power Laser Module.



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Your Snapmaker lets you turn your ideas into reality. You can engrave or cut on dozens of materials like paper, basswood, plywood, MDF, anodized aluminium, stainless steel, acrylic, and leather. Keep reading to learn about the recommended Luban settings for different materials.

If you wish to achieve optimal engraving or cutting results with other untested materials, you can utilize the **Material Test** feature to conduct your own tests.

This guide applies to the following types of the 10W High Power Laser Module.

- Acrylic
- MDF
- Leather
- Basswood
- Cardstock
- Coated Paper
- Corrugated Paper
- Glass
- Pinewood
- Stainless steel
- Anodized Aluminum

* Make sure you've set the best work origin or add a background for laser engraving and cutting.

How to Use the 10W Laser Module via Camera Capture – Snapmaker

How to Use the 10W Laser Module via Work Origin – Snapmaker

* The recommended settings might not apply to all situations. Take them as a reference, and you can fine-tune the parameters for your materials :)

* The 10W High Power Laser Module is the class 4 laser product. You should always operate it with the Enclosure developed by Snapmaker and wear the Laser Safety Goggles provided by Snapmaker. Meanwhile, the operator and bystanders should wear the Laser Safety Goggles provided by Snapmaker during the whole process.

Acrylic



Red Acrylic

For 10W High Power Laser Module

Engrave ✓

Cut ✓

	Thickness	Mode	Method	Power	Jog Speed (mm/mintute)	Work Speed (mm/mintute)	Number of Passes	Z Step per Pass
Cut	Cutting 2mm	Vector	On the path	100%	3000	180	1	0
	Cutting 3mm	Vector	On the path	100%	3000	120	1	0
	Cutting 5mm	Vector	On the path	100%	3000	100	1	0

Black Acrylic

For 10W High Power Laser Module

Engrave ✓

Cut ✓

	Thickness	Mode	Method	Power	Jog Speed (mm/mintute)	Work Speed (mm/mintute)	Number of Passes	Z Step per Pass
Cut	Cutting 2mm	Vector	On the path	100%	3000	180	1	0
	Cutting 3mm	Vector	On the path	100%	3000	120	1	0
	Cutting 5mm	Vector	On the path	100%	3000	100	1	0

MDF

MDF

For 10W High Power Laser Module

Engrave ✓

Cut ✓

		Mode	Method	Power	Jog Speed (mm/mintute)	Work Speed (mm/mintute)	Other Settings	
Engrave	Dot-filled Engraving	Greyscale	Fill	45%	3000	/	Dwell Time(ms/dot):5	
	Line-filled Engraving	Greyscale	Fill	25%	3000	800		
	Vector Engraving	Vector	On the path	50%	3000	2000		
	Thickness	Mode	Method	Power	Jog Speed (mm/mintute)	Work Speed (mm/mintute)	Number of Passes	Z Step per Pass
Cut	Cutting 2mm	Vector	On the path	100%	3000	600	1	0
	Cutting 3mm	Vector	On the path	100%	3000	240	1	0

Leather



Vegetable Tanned Leather

For 10W High Power Laser Module

Engrave ✓					Cut ✓			
		Mode	Method	Power	Jog Speed (mm/mintute)	Work Speed (mm/mintute)	Other Settings	
Engrave	Dot-filled Engraving	Greyscale	Fill	30%	3000	/	Dwell Time(ms/dot):5	
	Line-filled Engraving	Greyscale	Fill	40%	3000	3600		
	Thickness	Mode	Method	Power	Jog Speed (mm/mintute)	Work Speed (mm/mintute)	Number of Passes	Z Step per Pass
Cut	Cutting 1.5mm	Vector	On the path	80%	3000	1080	1	0
	Cutting 2mm	Vector	On the path	100%	3000	600	1	0

Crazy Horse Leather

For 10W High Power Laser Module

Engrave ✓					Cut ✓			
		Mode	Method	Power	Jog Speed (mm/mintute)	Work Speed (mm/mintute)	Other Settings	
Engrave	Dot-filled Engraving	Greyscale	Fill	30%	3000	/	Dwell Time(ms/dot):5	
	Line-filled Engraving	Greyscale	Fill	20%	3000	1200		
	Vector Engraving	Vector	On the path	50%	3000	3000	Number of Passes:2 Z Step per Pass(mm):1	
	Thickness	Mode	Method	Power	Jog Speed (mm/mintute)	Work Speed (mm/mintute)	Number of Passes	Z Step per Pass
Cut	Cutting 2mm	Vector	On the path	100%	3000	600	1	0

Basswood

Basswood

For 10W High Power Laser Module

Engrave ✓					Cut ✓			
		Mode	Method	Power	Jog Speed (mm/mintute)	Work Speed (mm/mintute)	Other Settings	
Engrave	Dot-filled Engraving	Greyscale	Fill	55%	3000	/	Dwell Time(ms/dot):5	
	Line-filled Engraving	Greyscale	Fill	30%	3000	800		
	Vector Engraving	Vector	On the path	50%	3000	1200	Number of Passes:1	
	Thickness	Mode	Method	Power	Jog Speed (mm/mintute)	Work Speed (mm/mintute)	Number of Passes	Z Step per Pass
Cut	1.5mm	Vector	On the path	100%	3000	600	1	0
	3mm	Vector	On the path	100%	3000	300	1	0
	5mm	Vector	On the path	100%	3000	180	1	0
	8mm	Vector	On the path	100%	3000	300	2	0



Cardstock

Cardstock		For 10W High Power Laser Module						
		Engrave ✓				Cut ✓		
		Mode	Method	Power	Jog Speed (mm/mintute)	Work Speed (mm/mintute)	Other Settings	
Engrave	Dot-filled Engraving	Greyscale	Fill	43%	3000	/	Dwell Time(ms/dot):5	
	Line-filled Engraving	Greyscale	Fill	25%	3000	1200		
	Vector Engraving	Vector	On the path	25%	3000	1200	Number of Passes:1	
	gsm	Mode	Method	Power	Jog Speed (mm/mintute)	Work Speed (mm/mintute)	Number of Passes	Z Step per Pass
Cut	200g	Vector	On the path	90%	3000	1800	1	0
	300g	Vector	On the path	100%	3000	1500	1	0
	300g	Vector	On the path	100%	3000	1200	1	0

Coated Paper

Coated Paper		For 10W High Power Laser Module						
		Engrave ✓				Cut ✓		
	gsm	Mode	Method	Power	Jog Speed (mm/mintute)	Work Speed (mm/mintute)	Number of Passes	Z Step per Pass
Cut	Cutting 200g	Vector	On the path	100%	3000	1800	1	0
	Cutting 300g	Vector	On the path	100%	3000	1800	1	0
	Cutting 350g	Vector	On the path	100%	3000	1500	1	0

Corrugated Paper

Corrugated Paper		For 10W High Power Laser Module						
		Engrave ✓				Cut ✓		
	Thickness	Mode	Method	Power	Jog Speed (mm/mintute)	Work Speed (mm/mintute)	Number of Passes	Z Step per Pass
Cut	Cutting 1.6mm	Vector	On the path	100%	3000	960	1	0
	Cutting 3mm	Vector	On the path	100%	3000	600	1	0

Glass



glass

For 10W High Power Laser Module

Engrave ✓

		Mode	Method	Power	Jog Speed (mm/mintute)	Work Speed (mm/mintute)	Other Settings
Engrave	Line-filled Engraving	Greyscale	Fill	100%	3000	400	
Please covered pigments or other things on surface of glass to make it not transparent							

Pinewood

Pinewood

For 10W High Power Laser Module

Engrave ✓ Cut ✓

		Mode	Method	Power	Jog Speed (mm/mintute)	Work Speed (mm/mintute)	Other Settings	
Engrave	Dot-filled Engraving	Greyscale	Fill	45%	3000	/	Dwell Time(ms/dot):5	
	Line-filled Engraving	Greyscale	Fill	45%	3000	800		
	Vector Engraving	Vector	On the path	90%	3000	3600		
	Thickness	Mode	Method	Power	Jog Speed (mm/mintute)	Work Speed (mm/mintute)	Number of Passes	Z Step per Pass
Cut	Cutting 4mm	Vector	On the path	100%	3000	300	1	0
	Cutting 8mm	Vector	On the path	100%	3000	800	2	0

Stainless steel

Stainless steel

For 10W High Power Laser Module

Engrave ✓

		Mode	Method	Power	Jog Speed (mm/mintute)	Work Speed (mm/mintute)	Other Settings	
Engrave	Line-filled Engraving	Greyscale	Fill	100%	3000	1800		
	Vector Engraving	Vector	On the path	100%	3000	1800	Number of Passes:1	

Anodized Aluminum



Engrave ✓

		Mode	Method	Power	Jog Speed (mm/mintute)	Work Speed (mm/mintute)	Other Settings
Engrave	Line-filled Engraving	Greyscale	Fill	100%	3000	3500	
	Vector Engraving	Vector	On the path	100%	3000	3500	Number of Passes:1

Feel free to contact us for more help at support@snapmaker.com. We are always available to help you out.

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